

| Question Number | Answer                                                                                                                                                                                               | Additional Guidance                                                                                                                                                                                              | Mark     |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1(a)(i)</b>  | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>• addition of {biuret reagent / base and copper sulfate} (1)</li> <li>• purple colour (1)</li> </ul> | <p>ignore heating<br/>accept any spelling which is close unless it is different word (eg burette)<br/>ignore extra detail unless contradictory</p> <p>be generous on colour (e.g. pink-purple, lilac, mauve)</p> | <b>2</b> |

| Question Number | Answer                                                                                                                                                                                                                                                         | Additional Guidance                                                                                                                                                                           | Mark     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1(a)(ii)</b> | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>• same mass of (food) sample (1)</li> <li>• same volume of water (for extraction) (1)</li> <li>• control of another aspect of extraction method (1)</li> </ul> | <p>do not accept volume of extract</p> <p>accept solvent</p> <p>e.g. temperature / length of time of extraction / type of filter paper<br/>do not accept conditions for filtrate storage.</p> | <b>3</b> |

| Question Number  | Answer                                                                                                                                                                                                                                                                                            | Additional Guidance                                                                                                                                                                                         | Mark     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1(a)(iii)</b> | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• A semi-quantitative test gives {an estimate / a range} of the concentration of a substance (1)</li> <li>• A quantitative test determines the exact concentration of a substance (1)</li> </ul> | <p>Allow involves an element of judgment such as comparison / subjective</p> <p>Allow gives an eg {absolute / precise / specific / accurate / valid} value / it is objective<br/>done using colorimeter</p> | <b>2</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional Guidance                                                                  | Mark     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|
| <b>1(a)(iv)</b> | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• (the diagram shows that) a darker colour = more protein (1)</li> <li>• the {solution / result} of a ({biuret / food}) test is compared with (the tubes in) the diagram (1)</li> <li>• therefore the concentration is estimated by deciding which of the (known) protein solutions in the diagram is closest to the (food test) result (1)</li> </ul> | <p>accept reverse</p> <p>accept just result (of test / experiment etc.) compared</p> | <b>2</b> |

| Question Number | Answer                                                                                                                                                                                                                                     | Additional Guidance | Mark                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------|
| <b>1(b)(i)</b>  | <p>An answer including the following steps :</p> <ul style="list-style-type: none"> <li>protein in milk <math>505 \times 6.38 = 3221.9 / 3222</math> (1)</li> <li>ratio = <math>20556 \div (3221.9 / 3222) = 6.4(:1)</math> (1)</li> </ul> |                     | <p>correct answer with no working gains 2 marks</p> <p><b>2</b></p> |

| Question Number | Answer                                                                                                            | Additional Guidance | Mark     |
|-----------------|-------------------------------------------------------------------------------------------------------------------|---------------------|----------|
| <b>1(b)(ii)</b> | <ul style="list-style-type: none"> <li>credit named example of an organic molecule containing nitrogen</li> </ul> |                     | <b>1</b> |